



KSI
KUMARAGURU
SCHOOL OF
INNOVATION

24ADC001- BUSINESS INTELLIGENCE



2026-2027
LAB RECORD

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

INDEX

Expt. No	Title of the Experiment	Page No	Total Marks	Faculty Signature
1.	Explore the interface and basic features of the BI tool			
2.	Load and visualize a sample dataset.			
3.	Import a dataset into the BI tool. And Cleanse data by handling missing values, outliers, and inconsistencies.			
4.	Transform data to suit BI reporting requirements and Design a dashboard with key performance indicators (KPIs).			
5.	Develop interactive dashboards for dynamic data exploration.			
6.	Integrate data from various sources for comprehensive analysis			
7.	Implement advanced chart types (treemaps, heatmaps, etc.).			
8.	Apply BI tools for forecasting and predictive analytics.			

Average marks

Faculty Signature

EXPT NO:1	EXPLORE THE INTERFACE AND BASIC FEATURES OF BI Tool
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is the purpose of Business Intelligence (BI) tools like Tableau and Power BI?

BI tools help analyze, visualize, and present data to support better business decision-making.

2. List any three common data sources that Tableau and Power BI can connect to.

- Excel Files
- CSV Files
- SQL Databases

3. What is a dashboard in BI, and how is it different from a single chart?

A dashboard combines multiple visualizations into one interactive view, while a single chart displays only one type of visualization.

4. Define the following terms:

- a) Visualization

The graphical representation of data using charts, graphs, and maps.

- b) KPI

A measurable value used to evaluate the performance of a business or process.

- c) Data Filter

A feature used to display only the required subset of data.

5. Why is it important to clean and transform data before visualization?

It removes errors, handles missing values, and improves the accuracy and reliability of the analysis.

6. What kind of chart would you use to analyze stock price trends over time?

A Line Chart.

7. Explain the role of slicers in Power BI and filters in Tableau.

Slicers and filters allow users to interactively view and analyze specific portions of data.

8. What does a calculated field or measure do in BI tools?

It performs calculations on existing data to create new values for analysis.

9. What are the advantages of using interactive dashboards for financial data analysis?

They provide real-time insights, enable quick comparisons, simplify trend analysis, and improve decision-making.

10. Which function would you use to find the absolute difference between predicted and actual stock prices?

The **ABS()** function.

Example: **ABS(Predicted Price - Close Price)**

IN-LAB EXERCISE:

Objective:

To understand the basic interface, data import process, and fundamental visualization capabilities of **Tableau** and **Power BI** using a stock prediction dataset, and to explore their usefulness in analyzing financial data.

Tools Required:

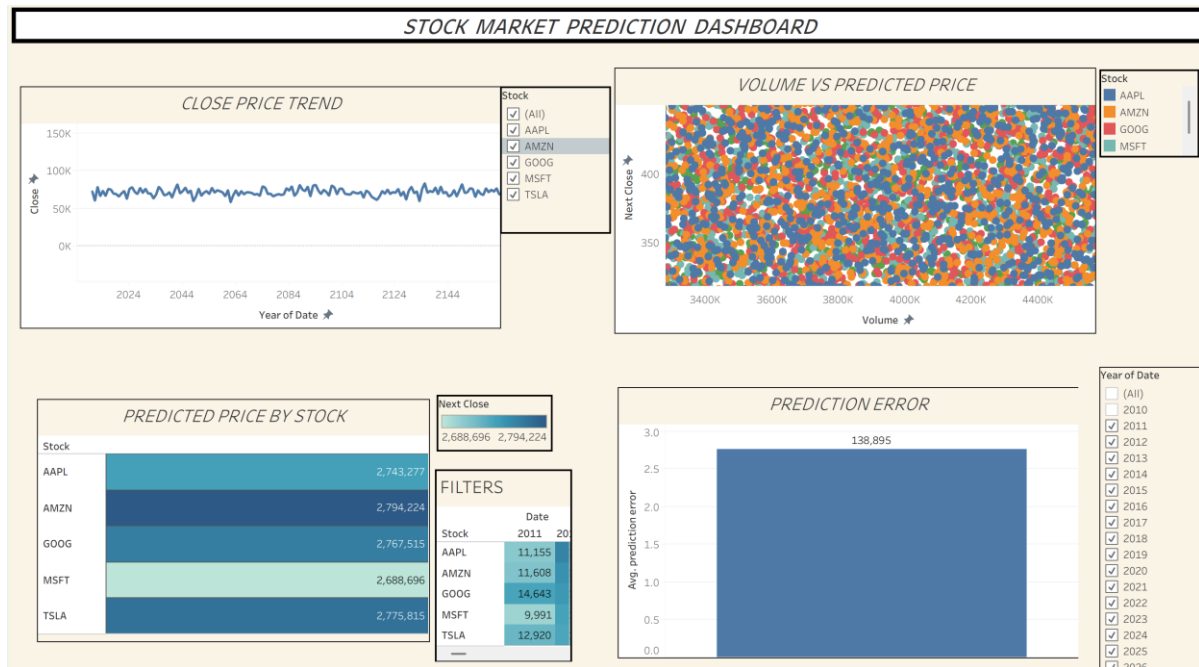
- Tableau Public / Tableau Desktop
- Power BI Desktop
- Sample Dataset (CSV/Excel) containing:
Date, Stock Name, Open, Close, High, Low, Volume, Predicted Price

Exercise Steps:

Part 1: Working with Tableau

1. **Launch Tableau** and connect to the stock prediction dataset.
2. **Explore the Interface:** ○ Understand components like **Data Pane**, **Sheets**, **Marks Card**, and **Dashboards**.
3. **Create Basic Visualizations:**
 - **Line Chart:** Show Close Price over time. ○ **Scatter Plot:** Plot Volume vs. Predicted Price.
 - **Highlight Table:** Display Predicted Price by stock.
4. **Apply Filters:**
 - Add filters for Stock Name and Date Range.
5. **Create Calculated Field:** ○ Calculate prediction error as $|\text{Predicted} - \text{Close}|$.
6. **Design a Basic Dashboard:**
 - Combine your charts into one interactive dashboard.

Screenshot:



Part 2: Working with Power BI

1. **Open Power BI Desktop** and import the same dataset.
2. **Explore the Interface:** ○ Familiarize with **Fields Pane**, **Visualizations Pane**, **Report View**, **Data View**.
3. **Create Basic Visualizations:**
 - **Line Chart:** Trend of Close Price over time.
 - **Bar Chart:** Compare Predicted Price vs. Close Price.
 - **Card (KPI):** Display average prediction error.
4. **Use Filters and Slicers:**
 - Filter data based on Stock Name and Date.
5. **Create a Simple Measure** using DAX:
 - Example: $\text{Prediction Error} = \text{ABS}(\text{Predicted} - \text{Close})$
6. **Build a Report Page:**
 - Arrange visuals in a single interactive page.

Screenshot:



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. Describe your experience with the Tableau interface. What components did you interact with?

The Tableau interface was easy to use. I worked with the **Data Pane, Sheets, Marks Card, Filters, and Dashboard** to create visualizations.

2. How did you import the dataset into Tableau and Power BI?

I selected **Connect to Data**, chose the CSV/Excel file, verified the data preview, and loaded the dataset into both Tableau and Power BI.

3. What types of visualizations did you create in both tools? List at least two per tool.

Tableau:

- Line Chart (Close Price over Time)
- Scatter Plot (Volume vs. Predicted Price)
- Highlight Table

Power BI:

- Line Chart (Close Price Trend)
- Bar Chart (Predicted Price vs. Close Price)
- KPI Card (Average Prediction Error)

4. How did you calculate the prediction error, and how was it displayed in your dashboard?

I calculated the prediction error using **ABS(Predicted Price - Close Price)** and displayed it as a KPI card and in charts for comparison.

5. Which filtering or slicing techniques did you apply, and what insights did they reveal?
I applied **Stock Name** and **Date** filters/slicers. They helped analyze individual stock performance and observe price trends over selected periods.
6. Compare the process of creating a line chart in Tableau vs. Power BI. What differences did you notice?
In Tableau, fields were dragged onto the Rows and Columns shelves. In Power BI, a Line Chart visual was selected and fields were assigned to the Axis and Values. Power BI uses a visualization pane, while Tableau uses shelves.
7. How did you organize the dashboard in Tableau and Power BI? What key elements did you include?
I arranged the charts neatly with filters/slicers at the top. The dashboard included a line chart, comparison chart, KPI card/highlight table, and interactive filters.
8. Which tool (Tableau or Power BI) did you find more user-friendly for financial data visualization, and why?
I found **Power BI** more user-friendly because of its simple interface, built-in DAX functions, and easy dashboard creation.
9. What challenges did you face while creating visualizations or dashboards?
I faced minor challenges in selecting appropriate chart types, creating calculated fields, and arranging visuals properly on the dashboard.
10. How can the knowledge gained from this lab be applied to real-world financial data analysis?
It can be used to monitor stock performance, compare predicted and actual prices, identify trends, track KPIs, and support data-driven investment decisions.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		